Recorded study metadata predicts the diagnosis in eight of nine public single-cell case-control comparisons

Short title: Study metadata and single-cell patient classification

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Abstract

Patient-level classifiers turn thousands of single cells into one score per donor, and routinely report cross-validated AUC above 0.95. That number cannot be read as a claim about disease if the variables recorded about how a donor was recruited, processed and sequenced also predict whether that donor is a case. We measure how far that is true. For each comparison we ask how well the recorded metadata alone predicts the diagnosis, and we judge the answer against a permutation null built from that comparison's own design matrix. We then run two permutation tests of the classifier itself. One shuffles the diagnosis labels freely. The other shuffles only within strata that share a collection configuration, so the link between diagnosis and collection survives while disease-specific structure does not. Where a quantity is undefined rather than small, we say so. We applied the checks to nine case-control comparisons from five public datasets (809 donors), spanning lupus, COVID-19, influenza, cytomegalovirus infection and a sepsis-versus-COVID-19 contrast. Recorded metadata alone predicts the diagnosis at cross-validated AUC 0.470 to 1.000, significantly so in eight of the nine; using collection variables alone, in four of the seven that record any. The exception is a cytomegalovirus cohort chosen in advance as a negative control: its metadata is at chance in all five seeds while its diagnosis classifier reaches AUC around 0.91. At the other extreme, a 21-donor influenza comparison has metadata that separates the labels exactly, leaving the restricted permutation nothing to shuffle. Between these ends the checks localise the problem. Two COVID-19 cohorts with similar overall metadata predictability, both around 0.8, are affected through different variables: sample quality in one, recruiting hospital in the other. None of these checks, alone or together, shows that a classifier's accuracy is biological.

Keywords: patient-level classification; single-cell RNA sequencing; study metadata; confounding; permutation tests; external validation; systemic lupus erythematosus; COVID-19

Author summary

Single-cell sequencing of a blood sample can be turned into a diagnostic score for one patient, and published scores often separate patients from healthy donors almost perfectly. Before trusting such a score, a clinician needs to know whether it reads the disease or reads the study. In most public datasets, patients and healthy donors were not recruited the same way. They may come from different hospitals, different years, or different laboratory runs, and a model can score well simply by recognising where a sample came from.

We measured how large this problem is across nine patient-versus-control

comparisons from five public datasets, covering lupus, COVID-19, influenza and cytomegalovirus infection. For each one we asked how well the recorded study details alone predict who is a patient. In eight of the nine, they predict it well. The exception is a cohort we chose in advance as the case that should come out clean, and it does.

We also show that the usual way of testing these models can miss the problem, because it destroys the very pattern it should be examining. Shuffling patient labels only within the same collection configuration asks a complementary question, and on one comparison the two tests disagree.

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1. Introduction

A clinician reading a single-cell paper is asked to believe a specific claim: that a blood sample, sequenced cell by cell and summarised into one number per patient, separates disease from health at an AUC of 0.95 or better. The relevant question is not whether the number is real. It is whether the number would survive a move to another hospital.

A growing family of methods competes to produce that one number per patient: pseudobulk aggregation, phenotype-oriented summaries, prototype networks, sparse gene panels, multicellular factor models, graph representations, and methods that first identify the disease-relevant cell population [6-11,32-35]. Frozen single-cell foundation models offer a convenient route - encode each cell once, average, train a small classifier on donors - though benchmarks report that such embeddings can retain batch structure and can lose to simple expression baselines [13,14]. Our concern is not which of these wins. It is that the comparison between them is made on cohorts where the winner may be decided by something other than biology.

That question is hard to answer in public data because cases and controls are usually not collected the same way. They may be recruited from different clinics, sequenced in different years, run on different chemistries, or processed in different batches. When that happens, a model can reach a high accuracy by learning the collection process. Batch-label confounding has been known to bias cross-validation for a decade [16], and a fully confounded single-cell design is not identifiable [17]. Adjusting for technical covariates does not fix it: regression removes the part of the data that design can predict, whether that part is technical, biological, or both [18,19]. Work in predictive modelling has shown the same thing from three directions - adjustment must be done inside training folds, confound regression can leak information to nonlinear learners, and residual confounding survives apparently successful correction [27-29]. Benchmark-side work raises the matching concern about how the datasets themselves are built [12,15].

None of this is in dispute. What has been missing is a number. How strongly does the recorded metadata predict the diagnosis in a given cohort? Which recorded variables carry that prediction? How does the answer compare with what the same variables would produce by chance? And when is the question answerable at all? Without those numbers, “this cohort is confounded” is an opinion, and “we adjusted for batch” cannot be checked.

A word on “recorded metadata”, because the distinction runs through the whole paper. The released tables mix three kinds of variable. Collection facts are study design in the strict sense. Demographics are case mix, and may also be genuine risk factors. Sample-quality summaries sit downstream of the assay, but can sit downstream of the disease too. Lumping them together and calling the result "study design" would overstate what a positive result means. We keep them apart, treat collection as primary, and report the headline count under both definitions.

This paper supplies the numbers for nine case-control comparisons drawn from five public datasets, and reports what they show. Our claims are deliberately narrow. We do **not** claim a general identifiability theorem for patient-level classification. The quantity we report is a linear association measure with stated blind spots. We do not claim that a cohort passing these checks yields biologically valid predictions, nor that one failing them yields none.

We claim three things. First, that the association between recorded metadata and diagnosis is graded and measurable, and varies across routinely reused cohorts by more than half the usable AUC scale. Second, that it can be traced to an interpretable group of variables, which matters because different groups call for different remedies and carry different interpretations. Third, that the permutation test normally used to validate these pipelines answers a different question from the one at issue; we give a comparison where the two answers differ, and we measure what the replacement test does and does not protect against.

These checks are complementary, not a standard. Each measures one thing, and we state the scope of every quantity where it is defined. Section 8 records what the approach cannot see.

2. Approach: what is being measured

2.1. The setting

Figure 1 sets out the situation in the notation of graphical causal models [36,37]. A recruitment process **R** assigns each donor a place in the recorded design **D** - site, batch, calendar period, chemistry, sample-quality profile - and is also associated with the diagnosis **Y**, because cases and controls are usually found through different routes. The measured cells **X** are affected by **D** through technical effects and by **Y** through biological ones. A classifier maps X to a prediction $\hat{Y}$.

Two further nodes matter in practice.

C stands for covariates that are consequences of the disease: severity, WHO score, comorbidity, medication, time since symptom onset. These sit on the path Y → C → X. They are not collection variables. Treating them as design variables converts "the disease is predictable" into "the design is predictable", and Section 4.1 quantifies how badly.

U stands for collection structure that was never recorded. Nothing in this paper can see U. That is a limit of the data rather than of the method, and Section 8 says so.

The graph was specified before the analysis and is released in machine-checkable `dagitty` syntax [38].

One thing the arrow between D and Y is not. We draw the association as induced by R rather than as a directed edge. Both directions occur in practice. A patient may be recruited at a referral centre because they are a patient, and a centre's case mix may determine which diagnoses appear in a batch. These checks measure the association. They do not tell you which way it runs.

2.2. Residual label variance

Let *y* be the centred diagnosis vector and **D** the recorded design matrix. Write

V_D = 1 − R^2(y ∼ D).

This is the share of case-control variation that the recorded design variables cannot explain in a linear model. It is a residual variance, not a Shannon or Fisher information: it is linear, and it depends on how D is coded. We report it with three things it needs in order to be readable.

First, cross-fitting. An in-sample R^2 inflates with the ratio of design columns to donors, so we compute V_D out-of-fold using five-fold cross-fitting. The correction is not cosmetic. Our CMV cohort has 89 design columns for 108 donors, so its in-sample R^2 is inflated to 0.532 and its in-sample V_D falls to 0.468, which would rank it the third-most affected of our nine comparisons. Its own permutation null sits at 0.496. The number was never below chance.

Second, the design matrix's own null. We shuffle the diagnosis 200 times and push each shuffled label through the same five folds, so the null is built from the same cross-fitted statistic as the observed value. This gives a one-sided p-value, p(V_D), for that specific design matrix at that specific donor count. A raw V_D cannot be compared across cohorts; its position inside its own null can. Observed statistic and null statistic are the same quantity by construction. The in-sample pair is released alongside, with its own null, and p(V_D) refers throughout to the cross-fitted pair.

Third, a classifier, and the test we actually rely on. We report the cross-validated AUC of a classifier using only the columns of D, fitted with penalised logistic regression and with a random forest. We call this the **design-only AUC**; the name refers to the design *matrix*, and which columns are in that matrix is stated every time - all recorded metadata, or the collection block alone.

This quantity, not V_D, carries the significance test. The reason is a limitation of V_D that only became visible once its null was matched to it. V_D is an *unpenalised* linear R^2, and cross-fitting an unpenalised fit loses power as the design matrix gets wider. The Ren COVID-19 comparison shows this. Its 26 one-hot columns on 182 donors leave the cross-fitted V_D at its ceiling, because the linear model explains nothing out of fold. A penalised logistic model on the very same matrix reaches AUC 0.82, and its own permutation null puts that at $p = 0.005$. Reporting only p(V_D) there would say "no association" about a comparison in which the recorded metadata plainly predicts the diagnosis.

So the primary test is a permutation null for the cross-fitted design-only AUC, with one frozen pipeline on both sides. V_D and p(V_D) are reported alongside as a linear-variance summary, and where the two disagree we say so.

What this does and does not cover. V_D looks at one arrow in Figure 1, the one between D and Y. Confounding of a classifier also requires that D affects X and that the classifier uses that part of X. A high design-only AUC is therefore necessary but not sufficient evidence that a reported diagnosis AUC is design-driven. A low one does not rule out confounding through unrecorded structure. Every use of V_D below is subject to this.

2.3. Two permutation tests that ask different questions

The usual complete-pipeline permutation test shuffles the diagnosis and repeats the whole analysis. It is a sound test of leakage, overfitting and pipeline implementation, and we keep it for that. But shuffling the diagnosis destroys the design-diagnosis association along with the biology. A significant result is therefore consistent with either explanation. It cannot test whether a classifier is exploiting the collection process, which is the question here.

We therefore add a **collection-preserving permutation**: diagnoses are shuffled only *within* strata that share a collection configuration - batch, pool, site and, where it varies, sequencing chemistry. This is the restricted permutation scheme used elsewhere in the confounding literature [25,30]. It holds the association between the diagnosis and those collection variables at its observed value while destroying disease-specific molecular structure.

What it preserves, and what it does not. The strata are built from collection variables only. The recorded metadata also contains sample-quality and demographic columns, and those are *not* conditioned on. So a significant result means the classifier scored above what these collection strata can explain. It does not mean the classifier beat everything recorded, and it is not evidence of a "design-independent component". If a sample-quality variable is still associated with the diagnosis inside a stratum, and drives expression, a classifier can pass this test with no disease effect present at all. We measured how often that happens (Section 3.4, Figure S1) rather than assuming it does not. Conversely, a non-significant result is a failure to reject under this restriction, not a demonstration that the accuracy is entirely collection-driven.

Conditioning on the full metadata instead would require a conditional randomisation scheme for a matrix containing continuous columns, which is a different and harder problem; we state the limit rather than overreach past it.

The two tests are not defined on the same data. The collection-preserving test needs at least one stratum containing both a case and a control. If every stratum has only one kind of donor, the shuffle is the identity, no null distribution exists, and the test returns nothing at all. This is not a large p-value or a small one. There is no number. Section 4.4 reports a comparison where exactly this happens.

2.4. Three verdicts, not two

We separate two failure modes that are easy to run together and that call for different responses.

A comparison **cannot be answered with these data** when the quantity has no meaning for it: no collection stratum holds both a case and a control, so the collection-preserving permutation cannot be formed. These are dropped from the main ordering and reported separately.

A discrepancy between a tuned and a frozen classifier is deliberately **not** used as a feasibility rule. Comparing two different algorithms across a cut-point would not establish that a comparison is unanswerable, and a tuned model outperforming a fixed one is not the same kind of problem as a design that separates the labels exactly. The frozen pipeline is the inferential statistic everywhere, because it is what both permutation nulls are built from. The tuned AUC sits beside it as a description. Their difference is released as a column, so a reader can judge pipeline sensitivity directly instead of through a threshold.

A comparison is **underpowered** when the quantity is well defined but unstable: fewer than 40 donors, or a minority class below 15. These stay in the ordering, flagged. Deleting them would hide the instability a reader needs to see. The gates themselves are deterministic - they depend on donor counts and stratum structure, not on the seed - but what an underpowered comparison then reports is not: Section 4.4 gives one whose permutation p crosses

0.05 in three seeds out of five.

Results

Sections 3 to 6 report results. Section 3 establishes how the checks behave on data with a known generating mechanism. Section 4 applies them to nine case-control comparisons. Section 5 asks whether a classifier actually uses the structure the checks detect. Section 6 states what each check can and cannot support.

3. How the checks behave in simulation

A simulation with one binary design variable, an additive design effect and constant noise would not show whether these checks behave when those choices are relaxed. We therefore crossed seven data-generating regimes with seven levels of design-diagnosis association and two outcome types, binary and continuous, each predicted and scored on its own scale: 19,600 simulated cohorts of 200 donors and 100 features (Figure 2). Nothing in the simulation is fitted to the real cohorts.

3.1. The null behaves

When design carries no information about the diagnosis, the checks must not fire. They do not. The fraction of runs with $p(V\_D) \leq 0.05$ is 0.025 to 0.065 across the seven regimes, which brackets the nominal 0.05 (Figure 2A). It rises at $\rho = 0.25$ and reaches one by $\rho = 0.50$ in every regime. This is the property the cross-fitted estimator and the per-cohort null were introduced to obtain; an in-sample estimator judged against an in-sample null does not have it.

3.2. Adjusting and restricting give different answers, in every regime

Two ways of handling design are in common use. Residualisation regresses the data on the design variables and analyses what is left. Restriction compares only donors collected under matching conditions. In the additive linear regime the two agree when design carries no diagnosis information and separate as it does (Figure 2B). At $\rho = 0$ the unadjusted, residualised, restricted and external AUCs are 0.806, 0.806, 0.788 and 0.817. At $\rho = 0.98$ they are 0.882, 0.536, 0.701 and 0.889. Residualisation has removed 0.35 AUC and restriction 0.18, on data where the biological effect never changed. At $\rho = 1$ restriction is not merely worse. It is undefined, because no stratum holds both labels.

The gap between the two opens in **every** regime (Figure 2C), so it is not an artefact of the linear setting.

The continuous-outcome arm is a genuine test of outcome type here, not a relabelling. Its cohorts are generated with a continuous latent outcome, predicted with ridge regression, and scored by concordance over pairs, which equals AUC when the outcome is binary and so puts both arms on one scale. The ordering survives: restriction beats residualisation at rho = 0.98 in all seven regimes for both outcome types. The magnitude does not. With a binary outcome the gap is 0.094 to 0.200; with a continuous one it is 0.013 to 0.038. So the qualitative conclusion does not depend on outcome type, and the quantitative one does - a simulation run only on continuous outcomes would have understated the problem by roughly a factor of five.

3.3. Two results we did not expect

What a rare diagnosis does, and what it does not. Two things travel together in an imbalanced cohort and are easy to confuse. One is prevalence itself. The other is that a collection boundary drawn at the middle of a population no longer lines up with a diagnosis drawn from its top fifth, so the design cannot separate the labels even when the two are perfectly linked in the latent model. We separate them with two arms. In `imbalanced`, both thresholds sit at the 80th percentile, so prevalence is the only thing that changed. In `imbalanced_offset`, the diagnosis sits at the 80th percentile while the collection boundary stays at the median. The result is that prevalence alone changes nothing. With thresholds matched, the imbalanced arm tracks the balanced one almost exactly: design-only AUC still reaches 1.000 at rho = 1 and V_D still falls to 0.000. It is the offset arm that blunts the checks, with design-only AUC saturating at 0.880 and V_D flooring at 0.728. The reassurance a rare diagnosis appears to give is therefore not about how rare it is. It is about whether the collection boundaries line up with the diagnosis, and a cohort can have that problem at any case rate.

Two of our own comparisons have minority classes of 10 and 11 donors, so this matters for reading them.

An external cohort collected the same way will confirm a design-driven model. In the base regime the external AUC does not fall as confounding rises. It *rises*, from 0.817 at $\rho = 0$ to 0.904 at $\rho = 1$, tracking the internal AUC almost exactly (Figure 2D). Only when the target is collected under a different design mechanism does the external estimate come apart, flattening at 0.816-0.829 across the whole range while the internal estimate climbs to 0.899.

The practical reading is uncomfortable. External validation checks design confounding only to the extent that the second cohort's collection process is genuinely independent of the first. Being a different dataset does not establish that. Two cohorts assembled by similar consortia, on similar platforms, through similar recruitment routes, can agree with each other and both be wrong. This sharpens Check 5 in Section 6 and is, in our reading, the most consequential thing the simulation adds.

3.4. What the collection-preserving test does not protect against

The collection-preserving permutation conditions on collection strata. The recorded metadata also holds sample-quality and demographic columns, which it does not condition on. If one of those is associated with the diagnosis *inside* a stratum, and also drives expression, then a classifier can beat the restricted null with no disease effect present anywhere. This is the test's main failure mode and it deserves a number rather than a caveat.

We simulated it directly. Cohorts of 200 donors are assigned to eight collection sites. A sample-quality variable predicts the diagnosis within each site, at a strength we vary. Expression depends on the site and on that variable, and on nothing else - there is no disease effect in the generating model at all. We then run both permutation tests exactly as they are run on the real cohorts (Figure S1, Table S1).

With no association at all between the quality variable and the diagnosis, both tests are calibrated: the collection-preserving permutation rejects in 6.7% of 300 runs (95% CI 4.4 to 10.1) and the free permutation in 7.7% (5.2 to 11.2), against a nominal 5%. As the within-site association strengthens, both rise together and neither is more protective than the other. At a mean within-stratum correlation of 0.29 the collection-preserving test rejects in 15.7% of runs and the free test in 17.3%; at 0.61 both reject in 89.3%. The classifier's own AUC rises from 0.501 to 0.650 over the same range, and there is no disease effect anywhere in the generating model.

The reading is that a significant collection-preserving result is evidence against the collection strata alone, and is not evidence of a disease-specific signal. Where a sample-quality or demographic block is itself associated with the diagnosis - which Table 1 reports for every comparison - that possibility has to stay open.

3.5. Residualisation loss without any confounding

Residualisation loss is often read as a measure of how contaminated a cohort was. It is not, and the size of the effect that has nothing to do with contamination can be measured. We generated cohorts in which the design is independent of the diagnosis by construction, with a real biological effect calibrated so the unadjusted AUC matches the CMV cohort's 0.845, and residualised on design matrices of increasing width. A second condition reuses the exact level sizes of the real CMV batch-by-pool crossing, which is far more ragged than equal blocks: 46 levels for 108 donors, 17 of them holding one person.

Loss appears without any confounding, it is large, and it is not monotone in width. At one to 32 design columns residualisation costs nothing - the mean loss is negative, between -0.065 and 0.001, because removing a little variance leaves the biological direction intact. Loss then peaks in the middle of the range, at 0.222 for 48 columns and 0.229 for 64, before falling again to 0.124 at 80 columns and 0.045 at 92, the width closest to the CMV cohort's 89. Run-to-run spread is wide throughout: the central 95% of runs at 92 columns spans -0.048 to 0.143. The ragged CMV-shaped layout, which realises 45 columns from the observed pool sizes, costs 0.091 on average (-0.024 to 0.228) - twice what balanced blocks of similar width cost, which is the effect we expected, but still well short of the 0.284 seen in the real cohort (Figure S2). We therefore report width and raggedness as a partial explanation and no more; Section 8 returns to the gap.

4. Nine case-control comparisons

4.1. Datasets, and what the recorded metadata contains

We assembled five public datasets [2,39-42] through the CELLxGENE Discover curation API and reduced each to one row per donor: log1p CPM pseudobulk expression, plus variables built from schema-guaranteed observation fields (`donor_id`, `disease`, `sex`, `development_stage`, `assay`, `tissue`) and cohort-specific collection variables recovered from the distributed object.

These three groups are not the same kind of thing, and we do not treat them as one.

- **Collection** (site, sub-study, sequencing chemistry, assay, batch, pool) is study design in the strict sense: it describes how the material was gathered and processed, and nothing about it is caused by the disease.
- **Demographic** (age, sex, ethnicity) is case mix. An imbalance here is a property of who was recruited, so it is a design fact; but age and sex are also genuine risk factors for most of these diagnoses, so an association with the label can be aetiological rather than procedural. The two readings cannot be separated from the data.
- **Sample quality** (cells per donor, mean UMI, genes per cell, mitochondrial fraction) is measured downstream of everything. It reflects the assay, but it can also reflect the disease, through cell composition, treatment or the state of the sample at collection. Section 4.1's mediator

result shows exactly how much damage that can do.

Because of this, **collection is the primary definition** wherever a result is read as a statement about study design, and the other two groups are reported beside it as prespecified sensitivity analyses. The union of all three is what we call the *recorded metadata*, and every claim about the union is worded as such. Section 4.2 gives the headline count under both definitions, because they differ.

Covariates that are consequences of the disease are excluded from all three groups by an explicit list: severity, outcome, comorbidity, medication, symptom timing, diagnosis fields, stage, grade, WHO score. These are the C node of Figure 1.

The exact raw columns and expanded design columns entering every group, for every comparison, are generated directly from the model-matrix construction code and released as Table S2. A machine check confirms that the released inventory reproduces the width of every model matrix actually fitted (34 of 34 blocks) and that no diagnosis label or label proxy appears in any of them.

This exclusion is not a formality, and we quantified what it is worth. In a simulation where design and diagnosis are completely unrelated, admitting a single disease-caused covariate to the design variables raises the design-only AUC from 0.488 to 0.793. It also flags the cohort as significantly affected in **100% of runs**, against 5.5% without it (Supplementary Figure S3, Supplementary Table S3). Residualising on the enlarged set then removes 0.12 AUC from a cohort in which nothing was confounded at all. A single mediator admitted by accident is enough to manufacture the entire finding.

Nine case-control comparisons were formed (Table 1): lupus versus healthy in GSE174188 (CD4 subset and all cells), COVID-19 versus healthy in three independent cohorts, influenza versus healthy and sepsis versus COVID-19 in COMBAT, and CMV-positive versus CMV-negative in the HIHA cohort. The five cohorts hold 809 distinct donors (261, 196, 120, 124 and 108). Comparisons drawn from the same cohort share donors, so the per-comparison counts in Table 1 do not sum to a donor total.

Comparison	Donors	Minority	Recorded: cols	AUC	p	V_D	p(V_D)	Collection: cols	AUC	p	Diagnosis AUC	p free	p coll.-pres.	Verdict
Influenza · COM-BAT	21	10	7	1.000	0.0050	0.000	0.0050	1	1.000	0.0050	0.273-0.582	0.2987-0.9660	—	**not estimable**
Sepsis vs COVID-19 · COM-BAT	111	11	7	0.972-0.992	0.0050	0.178-0.203	0.0050	1	0.984-0.984	0.0050	0.653-0.731	0.0240-0.1069	0.4506-0.9381	**underpowered**
SLE · GSE174188 (CD4)	261	99	12	0.903-0.904	0.0050	0.540-0.558	0.0050	3	0.759-0.768	0.0050	0.935-0.963	0.0010	0.0010	estimable
COVID-19 · COM-BAT	110	10	7	0.840-0.862	0.0050-0.0100	0.906-1.000	0.0050-1.0000	1	0.503-0.516	0.7413-0.8657	0.996-0.999	0.0010	0.0010	**underpowered**
SLE · GSE174188 (all cells)	261	99	9	0.847-0.851	0.0050	0.645-0.661	0.0050	n/a	n/a	n/a	0.972-0.987	0.0010	0.0010	estimable
COVID-19 · Ren	182	25	26	0.805-0.832	0.0050	0.847-1.000	0.0050-1.0000	19	0.733-0.750	0.0050	0.961-0.982	0.0010	0.0010	estimable
COVID-19 · Stephenson	115	29	8	0.795-0.806	0.0050	0.628-0.722	0.0050	2	0.549-0.565	0.0149-0.2338	0.909-0.939	0.0010	0.0010	estimable

Comparison	Donors	Minority	Recorded: cols	AUC	p	V_D	p(V_D)	Collection: cols	AUC	p	Diagnosis AUC	p free	p coll.-pres.	Verdict
COVID-19 · Ren (10x 5’ v2)	161	25	7	0.688-0.743	0.0050-0.0149	0.920-0.964	0.0050-0.0149	n/a	n/a	n/a	0.921-0.964	0.0010	0.0010	estimable
CMV · HIHA	108	45	89	0.470-0.501	0.3582-0.8657	1.000	1.0000	80	0.444-0.464	0.5224-0.9403	0.854-0.911	0.0010	0.0010	estimable

Table 1. The nine comparisons, ordered by how well the recorded metadata predicts the diagnosis. Generated from the screen output. Every comparison was run at five random seeds; each value is the range over those five, and a single value means all five agreed to the printed precision. "Minority" is the smaller of the two label groups. "Recorded" is all three variable groups together; "Collection" is the collection block alone, which is the study design in the strict sense, and n/a means the comparison records no collection variable that varies. Column counts are one-hot expanded columns, not source variables. AUCs are cross-fitted, and "Diagnosis AUC" is the frozen prespecified pipeline - the one both permutation tests are built from. Each p is one-sided against a permutation null built from that comparison's own design matrix. The two floors, 0.0050 and 0.0010, are 1/201 and 1/1001: the design-only AUC and V_D nulls use 200 permutations, the complete-pipeline tests 1,000. A dash means the collection-preserving permutation is undefined because no stratum holds both labels. Verdicts are read from the screen's own gates: **not estimable** when no collection stratum holds both labels or the metadata separates the labels exactly, **underpowered** when the minority group has fewer than 15 donors. The tuned nested-CV diagnosis AUC, the in-sample V_D and its null, and the individual per-seed values are in Tables S4 and S8.

4.2. A negative control chosen in advance

We chose the CMV cohort before computing anything, on its recorded structure alone. Its 108 donors were assayed on one chemistry and one suspension type, and the CMV status of a donor is a serological result rather than a recruitment criterion, so we expected the collection labels to be close to independent of who is positive. If these checks measure what they claim to measure, this cohort has to come out at the bottom.

The structure needs stating precisely, because the phrase “single-centre cohort” invites the wrong picture. This cohort is **not** one batch: the released metadata records 36 batch identifiers and 46 pool identifiers, and their crossing gives 46 strata, 17 of them holding a single donor. Neither is it donor-paired in any sense this analysis uses - each donor contributes one row with one label, and no within-donor contrast is formed anywhere in the paper. What makes it a negative control is not a simple design but a weak association between the collection labels and the diagnosis, which is exactly the quantity being screened.

It comes out at the bottom. Across five random seeds the design-only AUC for the full recorded set is 0.470 to 0.501, straddling chance, with permutation p 0.358 to 0.866 - not significant in any seed. The collection block alone gives 0.444 to 0.464, p 0.522 to 0.940, and sample quality 0.484 to 0.529, p 0.478 to 0.716. The same 108 donors support a diagnosis classifier at AUC 0.896 to 0.915, with both permutation tests at their 1/1001 resolution floor in every seed. That is the prediction, and it holds.

One qualification we report rather than bury. The demographic block alone reaches design-only AUC 0.611 to 0.625, and its permutation p ranges from 0.005 to 0.075 across the five seeds - significant in some seeds and not others, and not surviving correction across four blocks and nine comparisons. Sex and ethnicity are established correlates of cytomegalovirus seroprevalence in their own right, so an association there is as easily aetiology as recruitment. This is exactly the case the three-way split in Section 4.1 is for: it is a reason to read that block as case mix, and a reason not to let it stand for study design.

This is the falsifiable prediction in the paper, and it is why the CMV cohort is here. A method that flagged this cohort would be measuring donor count or design-matrix width rather than design-diagnosis association. Read without its own null, the in-sample V_D of 0.468 looks like a strong association - 89 columns fit 108 donors closely whatever the labels are. Against the matched null it is 0.468 versus a null mean of 0.496, p = 0.284 to 0.353: no association at all. This is the concrete case for building each statistic's null from the same design matrix rather than comparing it to a fixed reference.

4.3. A graded problem, not a binary defect

Ordered by the AUC from all recorded metadata, the nine comparisons run from 0.470 to 1.000 (Figure 3A, Table 1), with no gap that would justify a threshold. Positions are stable across seeds; the widest five-seed spread is 0.055. Eight of the nine have a significant association with the recorded metadata; the CMV negative control is the ninth.

Two comparisons show why the design-only AUC rather than V_D carries the test (Section 2.2). In COVID-19 Ren (26 design columns, 182 donors) and COVID-19 COMBAT (7 columns, 110 donors) the cross-fitted V_D is unstable across seeds, running from 0.847 to its ceiling of 1.000 in Ren and 0.906 to 1.000 in COMBAT. When it reaches the ceiling the matched null reaches it too, so p(V_D) is 1.000; in the other seeds the same comparison gives p(V_D) = 0.005. The design-only AUC on the very same matrices is stable by contrast, 0.805-0.832 and 0.840-0.862, significant in every seed (p = 0.005 and 0.005-0.010). Counting by p(V_D) alone would give six of nine rather than eight; we report both counts and use the better-powered one.

Under the primary, collection-only definition the count is smaller, and we give it rather than leave the union to stand for study design. Seven of the nine comparisons record any collection variable at all; in four of those seven the collection block alone predicts the diagnosis significantly, and two of those four are the comparisons that are not estimable or underpowered. Among comparisons that are both estimable and record collection variables, two of four are significant: lupus in GSE174188 CD4 (0.759-0.768) and COVID-19 in Ren (0.733-0.750). This difference is the point of separating the groups: much of what the union detects is carried by sample quality and case mix, which are not experimental design and which carry their own interpretations (Section 4.1).

Grouping the design variables shows that similar overall strength can come from different parts of the collection process (Figure 3B). The clearest case is a pair of COVID-19 cohorts of similar size and almost identical design-only AUC (Figure 3C):

- **Stephenson** (n = 115, recorded-metadata AUC 0.795-0.806). Sample-quality variables alone reach 0.808-0.853 (p = 0.005 in every seed), while collection variables reach only 0.549-0.565 and are not significant in every seed (p = 0.015-0.234). The metadata-diagnosis association is carried mainly by sample quality.
- **Ren** (n = 182, recorded-metadata AUC 0.805-0.832). The mirror image: collection variables reach 0.733-0.750 (p = 0.005 in every seed), while sample quality reaches only 0.617-0.632 and is not significant (p = 0.050-0.154). The association is carried mainly by recruiting hospital and sub-study.

A single number would call these two cohorts equally affected and would recommend the same fix. They need different ones. Harmonised preprocessing may help the first, but it would not address the site and sub-study imbalance in the second. This is also the direct answer to the objection that V_D depends on how the design matrix is coded. The objection is correct. The response is to report the grouping, not to hunt for a coding-free number.

The Ren cohort also shows how an incomplete set of design variables misleads. Our first pass recorded only sequencing chemistry as collection metadata, giving a collection AUC of 0.553 - apparently untroubled by site. Recovering the recorded hospital (12 centres, five of them case-only) and sub-study identifier (six source studies) raised that group to 0.750 and the full set from 0.758 to 0.818. The problem was there and invisible. These checks are only as good as the metadata they are given, and a reassuring result computed on thin metadata is weak evidence.

4.4. Where the two tests disagree, and where neither is defined

No number at all. The COMBAT influenza comparison has 21 donors and design variables that separate the labels exactly (design-only AUC 1.000, V_D 0.000). Every stratum holds one kind of donor (Figure S4), so the collection-preserving permutation has nothing to shuffle and returns nothing. The free permutation, on the same comparison, returns p = 0.299 to 0.966 across five seeds. Reported alone, that reads as a cohort with no detectable signal. The truth is that this comparison cannot separate the recorded metadata from the disease even in principle. We report it outside the main ordering.

Underpowered, and disagreeing. The COMBAT sepsis-versus-COVID-19 comparison (n = 111, minority class 11) is the case the collection-preserving test was added for. Its design variables predict the diagnosis almost perfectly, at 0.972 to 0.992 across seeds. The free permutation gives p = 0.024 to 0.107, significant at 0.05 in three of five seeds. The collection-preserving permutation gives p = 0.451 to 0.938, significant in none (Figure 3D). Under the conventional test this comparison would be reported as a validated classifier. Under the test that holds design fixed, its accuracy is entirely consistent with the collection process.

The frozen pipeline also scores 0.15 to 0.24 AUC below the tuned one here, the largest gap among the eight comparisons that return a number at all. That is not a feasibility failure. The frozen pipeline is the inferential statistic, and it is what both permutation tests are built from. We report the gap anyway: with 11 donors in the minority class, a reader should be able to see how much of the headline accuracy depends on model selection.

4.5. What this ordering does not show

The ordering says how well recorded metadata predicts the diagnosis. It does not show that any particular published classifier is design-driven, because it does not test whether the classifier uses the design-related part of X - the $D \to X \to \hat{Y}$ path in Figure 1. Sections 5.2 to 5.4 test that path directly in the two lupus cohorts, and reach a more equivocal answer than the ordering alone would suggest.

5. Does the classifier actually use it? Two lupus cohorts

The ordering measures the D-Y arrow. This section follows the $D \to X \to \hat{Y}$ path in the two lupus cohorts for which we hold complete donor design metadata, using frozen Geneformer [4] embeddings and two expression pseudobulk representations under one fold-contained pipeline. GSE285773 [3] serves as the independent transfer source. Figure 4A shows how cases and controls are distributed across the recorded design in both cohorts; Figure 4B shows how well the design alone predicts the diagnosis.

5.1. The representations carry batch information

In GSE174188 the three representations classified disease at AUC 0.982, 0.977 and 0.975. The same representations identified processing wave, one against the rest, at 0.9997, 0.999 and 0.997. In GSE135779 disease AUCs were 0.870, 0.933 and 0.945, while the best single-batch AUCs were 0.993, 0.961 and 0.961 (Figure 5A).

This shows that the representations contain batch information. It shows nothing more. A representation that encodes batch can still carry disease information, and a batch-aware representation need not produce a batch-driven decision rule. Batch predictability is therefore not a validity check on its own, and we report it below only with that scope attached.

5.2. Containing batch information is not the same as using it

GSE135779 provides the counterexample directly. The best one-against-rest batch AUC was 0.993 for Geneformer and 0.961 for both pseudobulk representations, which is close to perfect batch recognition. Yet batch alone predicted disease at AUC 0.499, and fold-contained batch residualisation changed disease AUC by +0.002, −0.002 and −0.009 (Figure 5B, 6C). The information was there, and removing it changed nothing measurable at this sample size - the three shifts are well inside the split-to-split spread, so we can say the classifier's dependence on batch is small, not that it is zero. That is enough to break the inference. Going from "the representation knows the batch" to "the classifier uses the batch" does not hold on these data.

5.3. Residualisation and restriction answer different questions

Restriction asks whether the separation survives among donors collected under overlapping conditions. Residualisation deletes the part of the data that a chosen set of design variables can predict. When design and diagnosis are collinear these are not the same question, and we compare them to show they are not interchangeable, not to rank them.

In GSE174188, ridge residualisation on processing-wave fractions reduced the three representations from 0.982, 0.977 and 0.975 to 0.714, 0.747 and 0.735. Restriction to the one large near-balanced wave, compared against size- and label-matched random donor subsets, cost only 0.014 to 0.049 (Figure 6B). The paired difference between the two losses was 0.186 to 0.255 across six comparisons, with every descriptive interval above zero.

In GSE135779, removing the single all-case batch left 36 donors with both labels in every remaining batch and changed matched performance by at most 0.012. Five ways of removing the recorded variables give five different answers for the same cohort (Figure 6A). Projecting out the complete set with ridge collapsed all three representations to 0.516, 0.522 and 0.520. A random-forest residualiser removed less (down to 0.748, 0.787 and 0.795), and overlap weighting and batch location adjustment less still (retaining 0.859 to 0.937). Every full-set removal took away far more than the batch-only control.

The algebra explains this without needing a technical story. When design predicts the diagnosis, the part of the data that design can predict necessarily includes diagnosis-related variation. Removing it removes disease signal regardless of where that signal came from. A large loss after residualisation is therefore **not** evidence that the original AUC was spurious. It is equally consistent with removing a shortcut and with removing real biology, and the size of the loss does not distinguish them.

A negative control settles part of what residualisation loss actually measures. In the CMV cohort no association is detectable between the full recorded set and the diagnosis: design-only AUC 0.483 under this pipeline, p(V_D) 0.284 to 0.353 across five seeds. Residualising its representations on those variables still costs 0.282 AUC, from 0.845 down to 0.563 (Figure 6C). There is nothing there to remove, and removing it costs a quarter of the separation.

Our first explanation was arithmetic: 89 one-hot columns for 108 donors, so the projection takes out most of any sample. We tested that rather than asserting it. We ran a simulation in which the design is independent of the diagnosis by construction, with the unadjusted AUC matched to this cohort. A balanced design matrix of the same width (92 columns) removes only 0.045. The loss is also not monotone in width: it peaks at 0.229 for 64 columns and falls away on either side. One built to the same ragged shape as the real batch-by-pool crossing - many levels holding one or two donors - removes 0.091, twice what balanced blocks of similar width remove (Figure S2). Width and raggedness account for part of the 0.282 and not for all of it. So the claim we can support is narrower than the one we first wrote: residualisation loss depends on properties of the design matrix that have nothing to do with confounding, and its size therefore cannot be read as a measure of technical contamination. We cannot fully account for this particular cohort's loss, and we report the gap rather than close it with a story.

It is tempting to read the pair of values from residualisation and restriction as a range bounding the true biological effect. It is not one. There is no proof that the two estimates bracket any underlying quantity. They are two estimates made under two different sets of assumptions, and their disagreement is informative precisely because neither is identified.

5.4. Cell-type composition shows the same pattern

Aggregating the same GSE174188 CD4 cells into naive, effector-memory, regulatory and unassigned fractions gave disease AUC 0.899 to 0.915. The same four-number summaries predicted processing wave at 0.803 to 0.872. Wave residualisation reduced them to 0.689 to 0.709 (Figure S5A), and within-wave restriction against matched subsets cost 0.027 to 0.062, with every interval crossing zero (Figure S5B). A four-dimensional biological summary therefore reproduces the whole pattern. That places the problem in the cohort rather than in foundation-model embeddings, and is consistent with composition being a strong patient stratification baseline in its own right [26].

5.5. Transfer to an independent cohort does not fix the problem

Restricting the training set to one processing wave is the obvious remedy: train where the design is uniform, then test elsewhere. It does not work. Training on all 261 GSE174188 donors and testing in the independent 26-donor GSE285773 cohort gave 0.900, 0.944 and 0.969. Restricting the source to the dominant wave lowered every one of them, and did no better than a random subset of the same size (Figure 7A). Within GSE174188, training on waves 2 and 3 and testing on wave 4 gave 0.842, 0.811 and 0.806, and the reverse direction 0.919, 0.926 and 0.821. More varied source donors transferred better than the apparently cleaner restricted source.

Transfer in the other direction, from the 26-donor cohort to the 261-donor target under strict source-only rules, gave 0.884 [0.843-0.920], 0.919 [0.884-0.946] and 0.926 [0.895-0.953]. Both pseudobulk representations beat Geneformer after paired DeLong testing [23] and Benjamini-Hochberg adjustment

[24] (q = 0.0151, q = 0.000603).

Whether a more expressive donor-level aggregator changes this is a separate question from the one this paper asks, and we report it in the supplement rather than the main text. Briefly: three learned pooling methods on the same frozen embeddings never beat ordinary mean pooling, and every significant paired difference was negative (Figure S6, Table S5). Adding capacity fitted the source without producing a representation that travels.

6. Five checks, and what each one can show

These are not a validity standard. We have not shown that they are sufficient, that they are minimal, or that passing them licenses a causal claim. They are five checks that look at different arrows in Figure 1 and that fail in different ways. Their value is that they can disagree with each other.

#	Check	Shows	Does **not** show
1	Design-only AUC by variable group, cross-fitted V_D, $p(V_D)$ against the cohort's own null, overlap counts	Whether recorded design predicts the diagnosis, and which variables do	That the classifier uses design; anything about unrecorded structure

#	Check	Shows	Does **not** show
2	Complete-pipeline permutation, both free **and** collection-preserving	Free: pipeline validity, no leakage. Collection-preserving: that accuracy beats what the collection strata alone can produce	Collection-preserving conditions on collection variables only, so it does not cover sample-quality or demographic association inside a stratum (Section 3.4); neither is defined when no stratum holds both labels; neither identifies a mechanism
3	How well the representation predicts design	That the representation contains collection information	That the diagnosis decision rule uses it — Section 5.2 is a counterexample

#	Check	Shows	Does **not** show
4	Residualisation **paired with** restriction and matched donor controls	How far two non-equivalent adjustments disagree	Which adjustment is right; bounds on a biological effect
5	Source-only external target	Whether patient ranking survives a change of collection process	Calibration or clinical usefulness; and it carries much less weight when the target was collected the same way, since a shared bias reproduces rather than cancels (Section 3.3)

Checks 1 and 2 cost minutes of CPU on a donor-level table and are the ones we would run first on any new cohort. Check 5 is the only one that speaks to transport, and no combination of the internal checks substitutes for it.

What a clinician should take from this. A published patient-level classifier is worth taking seriously as a candidate diagnostic when three things are reported together. The first is the design-only AUC for the development cohort. The second is both permutation tests, or a statement that the collection-preserving one is undefined. The third is an external result on a cohort collected through a different route. If the paper reports only a cross-validated AUC, the number is uninterpretable - not wrong, but not yet a claim about disease. If it reports an external validation on a cohort assembled the same way as the first, that still demonstrates reproducibility, but it does not rule out a bias shared by both.

Applied to our nine comparisons, the checks give three groups rather than a pass and a fail. In the first, no association with the recorded metadata is detectable, and internal accuracy is not undermined by anything we can see - which is weaker than saying it is biological, since unrecorded structure remains possible; only CMV is here. In the second, the metadata predicts the diagnosis strongly and yet the collection-preserving permutation is still beaten, so the accuracy exceeds what the collection strata alone explain; both lupus comparisons and all three COVID-19 cohorts are here. In the third, the collection-preserving permutation is either not beaten or not defined, and no internal analysis can settle the question; sepsis-versus-COVID-19 and influenza are here. Only the third group is a stop condition, and it is a statement about the cohort, not about the method being evaluated.

6.1. Three cohorts taken through the tree step by step

Figure 8 traces four comparisons through the decision tree with the evidence for each branch stated. Three are given in full here; the fourth, sepsis versus COVID-19, is described in Section 4.4. The strata and the design matrix are both imported from the screen rather than rebuilt for the walkthrough, so Q2 and Q4 ask about the same partition and the same columns.

COMBAT influenza (n = 21) stops at Q0. Its stratum variable is the recording institute, which gives two strata, *neither of which contains both a case and a control* (Figure S4). The collection-preserving permutation therefore has nothing to shuffle and is undefined. The comparison therefore cannot be answered with these data and no route in the tree is available. The frozen pipeline also scores far below the tuned one here - 0.27 to 0.58 against 0.98 to 1.00 across seeds. That gap is reported but is not a gate anywhere in this paper, and it is not what stops the trace. The stopping condition is the absence of a stratum holding both labels. Note what the free permutation test alone would have said: $p = 0.299$ to 0.966 across the five seeds, which reads as “no signal” and is the wrong description.

CMV HIHA (n = 108) reaches Route A. The collection-preserving permutation is defined - 46 strata from batch and pool, 21 of them containing both labels - and both permutation tests reject at the 1/1001 floor. At Q3 the association with the recorded metadata is weak (design-only AUC 0.483 under the walkthrough’s pipeline, 0.470 to 0.501 in the main screen, not significant against its own permutation null in any seed), so the comparison takes the adjustment route without reaching the restriction questions. It is the only one of our nine comparisons that does.

COVID-19 Ren (n = 182) reaches Route B2. Both permutation tests are beaten. The association with the recorded metadata is strong: design-only AUC 0.840 under the walkthrough’s fixed-penalty pipeline and 0.805 to 0.832 under the main one, both far above chance. Mixed strata exist, so Q4 passes and the comparison reaches the residualisation-versus-restriction question. Residualising the 26-column design matrix costs 0.100, from 0.948 down to 0.849 AUC. Restriction gives the opposite sign: within the one stratum large enough to cross-validate, AUC is 0.972 against 0.749 in size- and composition-matched random subsets. The two adjustments disagree in direction. That is

Route B2, and under the revised tree the trace stops there, with two estimates reported and no bound claimed.

The walkthrough also exposes a weakness in the tree itself. At the stratum definition the collection-preserving permutation uses, restriction is barely available in these cohorts. CMV has 21 label-mixed strata but *none* with five donors in each class. COVID-19 Ren has exactly one usable stratum, of 18 donors, whose matched-subset comparison carries a standard deviation of 0.117. Route B is drawn as though restriction were a practical alternative to residualisation. On real cohorts with realistically fine strata it frequently is not, and the honest output in those cases is Route C - redesign - rather than a restricted estimate computed on eighteen donors. We state this rather than loosening the stratum definition until Route B becomes available, which would make the tree unfalsifiable.

Discussion

7. What this means for reading a published classifier

Design-diagnosis association in public single-cell cohorts is measurable, graded, and traceable to specific variables. Across nine comparisons from five widely reused cohorts it runs from a design-only AUC of 0.470 to 1.000, which is more than half the usable scale, and a cohort's position is stable across random seeds. The *raw* residual label variance, by contrast, is not comparable across cohorts without its own null. That matters in practice: an in-sample R^2 ranks the least affected cohort in our set as the third most affected, purely because its design matrix is wide relative to its donor count.

Grouping the design variables matters more than the single number. Two

COVID-19 cohorts whose design-only AUCs differ by 0.02 are affected through different variables, one through library quality and one through recruiting hospital. Restriction to balanced strata is available for one and not the other. A quality-driven association may be partly correctable by harmonised preprocessing; a hospital-driven one is not. Reporting one number hides the only part of the answer a reader can act on.

The disagreement between the two permutation tests is, in our view, the finding with the widest reach beyond this dataset. Complete-pipeline label permutation is standard practice and is a good test of what it tests. But it cannot test whether a classifier exploits the collection process, because it destroys that process as part of the shuffle. On the sepsis-versus-COVID-19 comparison the two tests reach opposite conclusions, and the conventional one is the permissive one. Any study that uses label permutation to argue that a patient-level classifier is valid should report the collection-preserving version alongside it, or state that no mixed stratum exists and that version is undefined. That statement is itself the more informative outcome.

The replacement test has a failure mode of its own, and we would rather publish the number than the caveat. It conditions on collection strata, not on the whole recorded matrix, so a sample-quality variable that stays associated with the diagnosis inside a stratum can carry a classifier past it with no disease effect present. Section 3.4 measures how often. A significant collection-preserving result therefore rules out one specific explanation - these strata - and not the general one.

The simulation adds a caution about external validation that we did not anticipate. When the external cohort shares the source's collection mechanism, external AUC rises with confounding rather than falling. An external result is evidence about transport only in proportion to how independent the second collection process really is, and being a different accession number does not establish independence.

For representation research the implication is narrower than a leaderboard result but harder to work around. In our lupus data pseudobulk beats frozen embeddings in strict source-only transfer, and added pooling capacity does not change that. The more durable point is that internal accuracy in these cohorts is not a measurement of the property that matters, and no amount of model development fixes a cohort in which design predicts the diagnosis. That is a recruitment and data-release problem. The fix is to record and distribute collection metadata, and to build cohorts in which cases and controls appear together within batches - not to build a better encoder.

None of these results supports a diagnostic claim. The external AUCs we report rank 261 or 26 donors and are not calibrated probabilities. What the analysis supports is a reporting discipline. A patient-level classifier should be published with three things. First, the design-only AUC of its development cohort **by variable group**, so a reader can see whether the association runs through collection, case mix or sample quality. Second, both permutation tests, or a plain statement that the collection-preserving one is undefined. Third, an external result on a cohort collected through a different route, fitted on the source cohort alone. None of that requires a new method. It requires releasing the metadata and running two cheap checks.

8. Limitations

These checks see only recorded design. Unmeasured collection structure - referral patterns, institutional treatment protocols, sample handling not captured in released metadata - is invisible to every quantity we report. A low design-only AUC is therefore weak evidence of an unaffected cohort rather than strong evidence of one. The Ren cohort shows the same failure in its recorded form: our first set of design variables missed the recruiting hospital entirely and reported a collection-group AUC of 0.553 where the completed set gives 0.750.

The collection-preserving permutation conditions on a subset of the recorded metadata. Its strata come from collection variables; sample-quality and demographic columns are in the design matrix but not in the strata. Section 3.4 measures the consequence directly: with a sample-quality variable still associated with the diagnosis inside a stratum and driving expression, the test rejects far above nominal with no disease effect anywhere in the generating model. A significant collection-preserving result is therefore evidence against the collection strata, not evidence of a disease-specific signal, and it is not a demonstration that a "design-independent component" exists. Building a null that conditions on the full matrix, including its continuous columns, would require a conditional randomisation scheme we do not develop here. That is the single largest methodological gap in this paper.

The three variable groups are not equivalent, and only one is study design in the strict sense. Demographics can be case mix or genuine risk; sample-quality summaries sit downstream of the assay but can also sit downstream of the disease. We therefore report the collection block separately everywhere and treat it as primary for any claim worded as being about study design. The headline count differs between the two definitions, and both are given.

V_D is linear by construction and depends on how the design matrix is coded.

We reduce but do not remove this by reporting nonlinear design-only classifiers, the variable grouping, and the design matrix's own null. The nonlinear arms use two tree ensembles and are not an exhaustive comparison of adjustment algorithms.

We cannot fully account for the CMV residualisation loss. Section 3.5 shows that design-matrix width and raggedness produce a real loss with no confounding present, but less than the 0.282 observed. Something about that cohort's design matrix that we have not isolated accounts for the remainder.

All nine comparisons were run at five seeds, and every value we quote is the range over those five. Three comparisons have fewer than 15 donors in the minority class. We flag rather than exclude them, but no conclusion should rest on them.

The nine comparisons come from five datasets and are **not independent**. Comparisons drawn from the same dataset share donors, so the count of eight in nine is a count of comparisons, not of independent replications.

Matched random subsets separate sample-size loss from restriction without identifying a causal technical effect, so the restriction results show non-attributability rather than that retained signal is biological. Our coverage is four diseases from five cohorts, all peripheral blood, all public, all curated under one schema. Tissue-resident and non-blood settings are untested. Geneformer pretraining overlap with individual benchmark cells was not tested.

Finally, these are checks. They order cohorts and point at variable groups. They do not identify a causal effect, and we make no claim that they do.

Methods

9. Materials and methods

9.1. Cohorts and the unit of analysis

Seven public datasets were used, in two groups. Table S6 is the registry. It gives, for each dataset, the identifier, the donor count reported by the source, and the donor, case and control counts actually modelled - these differ, because donors below the minimum cell count or carrying a label outside the contrast are dropped. Five entries were verified against the CELLxGENE Discover API on 2026-09-07; the two used only in Section 5 come from GEO and are marked as not API-verified.

Five datasets enter the main screen, all obtained through the CELLxGENE Discover curation API: the CD4-positive alpha-beta T-cell subset of GSE174188 (261 donors; 162 cases, 99 controls) [2], the Ren COVID-19 atlas (196 donors) [39], the Stephenson COVID-19 cohort (120 donors) [40], the COMBAT consortium (124 donors) [41] and the HIHA cytomegalovirus cohort (108 donors) [42]. Nine case-control comparisons are formed from these five, so the comparisons are not independent of one another; Table 1 gives the donor overlap.

Two further lupus datasets are used only in the downstream analyses of Section 5, not in the screen: GSE135779 (44 childhood donors; 33 cases, 11 controls) [1] and GSE285773 CD4-positive T cells (26 donors; 16 cases, 10 controls) [3], the latter as the independent transfer target.

Registry entries, dataset identifiers, donor counts and download sizes were verified against the API on 2026-09-07 and are released in `cohorts/registry.yaml`.

The donor was the independent unit in every supervised analysis. Cells from one donor never crossed a training/test partition.

9.2. Donor-level tables and the design variables

Each cohort was streamed from its h5ad in 100,000-cell chunks and reduced to one row per donor. Expression was aggregated to log1p CPM pseudobulk. The layer used for aggregation was chosen by sniffing for integer counts rather than assuming a layer name.

Design variables were drawn from schema-guaranteed observation fields (`donor_id`, `disease`, `sex`, `development_stage`, `assay`, `tissue`) plus cohort-specific collection variables detected by pattern from the distributed object. Detection patterns cover batch, pool, run, lane, chip, site, centre, city, region, province, hospital, clinic, study, sub-study, source, dataset and donor source. Ages given as strings ("25-year-old stage", "fifth decade stage", "90 year-old and over stage") were parsed to years.

Covariates that are consequences of the disease were excluded by an explicit blocklist matching severity, outcome, comorbidity, medication, sample timing, symptom, diagnosis, stage, grade and WHO score. Supplementary Table S2 lists, for every cohort, which columns entered the design matrix, in which group, and which were excluded and why.

Variables were assigned to three groups. **Sample quality**: cells per donor, mean UMI per cell, mean genes per cell, aggregate mitochondrial percentage, and their logs. **Demographic**: age in years, sex, ethnicity. **Collection**: every detected `batch__*` variable, assay and suspension type.

9.3. Residual label variance

For centred diagnosis vector y and recorded design matrix **D** including an intercept, V_D = 1 − R^2(y ~ D). Two versions are reported. The in-sample version uses ordinary least squares on all donors and is reported for comparison. The cross-fitted version replaces the fitted values with out-of-fold predictions from five-fold `KFold` with shuffling, seeded by the run seed, and is the version used everywhere a number is interpreted. Negative R^2 values were clipped at zero before subtraction.

9.4. The design matrix's own permutation null

For each design matrix, the diagnosis vector was permuted 200 times. Each permuted label was pushed through the **same five folds** used for the observed value, so the null is a distribution of the cross-fitted statistic, not of a different one. The reported p-value is one-sided, (1 + #{null ≤ observed}) / (200 + 1), and answers: how often does a random diagnosis leave at most as much unexplained variation as the observed one? The null mean and its 2.5th percentile are reported alongside.

The design-only AUC is tested the same way, with 200 permutations of the diagnosis through the identical frozen pipeline (`p_design_auc` in Table S4); that is the primary test, for the reason given in Section 2.2. The in-sample V_D is released with its own in-sample null, as `V_D_insample` and `p_V_D_insample` in Table S4. The two are separate quantities and are never mixed: p(V_D) always refers to the cross-fitted pair. This null is what makes either version readable, because its location depends on the number of design columns relative to donors.

9.5. Design-only and diagnosis classifiers

Numeric design variables were median-imputed and standardised inside each outer training fold; categorical variables were most-frequent-imputed and one-hot encoded, with unseen held-out levels ignored. Every grid and fixed setting is listed in Table S7. All imputation, encoding, scaling, feature selection and regularisation choices were fitted on training donors only.

The main classifier was balanced logistic regression (liblinear, `max_iter` 20,000), with inverse regularisation strength selected from 10^{-4} to 10^{4} in decade steps by five-fold resampling inside each outer training set. Internal analyses used 20 repeated stratified five-fold donor splits with integer seeds 20260801-20260820. Out-of-fold predictions were pooled within each repeat; we report the mean AUC and the 2.5th-97.5th percentile of repeat-level AUCs. That range is a descriptive split-sensitivity range. It is not a confidence interval, and it does not treat correlated folds or repeats as independent [31].

Nonlinear design-only arms. We used two tree ensembles rather than a wider search because the question is whether a *low-dimensional* design-diagnosis relationship is being missed by a linear model, not which learner maximises AUC. Random forests used 256 trees, maximum depth 4, minimum leaf size 3. Histogram gradient boosting used 150 iterations, learning rate 0.05, seven terminal leaves, minimum leaf size 5. Both used fixed hyperparameters, fitted inside every outer training fold, and were not selected against held-out donors. Reporting them with fixed settings is deliberate: a tuned nonlinear arm would not be comparable with the tuned linear one under the same permutation null.

9.6. The two permutation tests

Both permutation tests and the observed statistic they are compared against run one identical frozen pipeline: standardisation, PCA to 50 components (or fewer when donors are limiting), balanced logistic regression at fixed inverse regularisation 1.0, and a single cross-fitting repeat. Freezing is necessary: comparing a tuned observed statistic against an untuned null biases p-values downward.

The **free** test permutes the diagnosis vector without restriction. The **collection-preserving** test permutes it only within collection strata. A stratum is the interaction of every detected `batch__*` variable with assay, where assay varies. Table 1 names the stratum variables for every comparison, and the screen writes them into its own output (`strata_definition`) so the name and the implementation cannot drift apart.

The conditioning set is collection variables only. Sample-quality and demographic columns are in the design matrix but not in the strata, so this test conditions on a subset of the recorded metadata. Section 3.4 measures the consequence: with a sample-quality variable still associated with the diagnosis inside a stratum and driving expression, the test rejects at a rate well above nominal even though the generating model contains no disease effect. Conditioning on the full matrix would need a conditional randomisation scheme for continuous columns, which we do not attempt here.

Where a cohort records no collection variable at all, the fallback is tertiles of log cells per donor. That is a sample-quality variable, so for those comparisons the test is a sample-quality-preserving permutation and is labelled as such rather than being folded in silently.

A permutation contributes to the collection-preserving null only if at least one stratum contains more than one donor and both labels; if no permutation qualifies, the p-value is undefined and reported as such rather than as a number. Main runs used 1,000 permutations, so the smallest value it can return is $1/1001 \approx 0.0010$. The V_D and design-only AUC nulls use 200, so their floor is $1/201 \approx 0.0050$. Every seed used the same counts; the two floors in Table 1 are these two tests, not two different runs.

9.7. Feasibility and power checks

A comparison is reported as **not answerable with these data** if the collection-preserving permutation has no qualifying stratum, or if the design-only AUC reaches 1.000, which is the complete-collinearity case. It is reported as **underpowered**, and kept in the main ordering with a flag, if it has fewer than 40 donors or fewer than 15 in the minority class. The two are separated because they call for different responses: the first cannot be fixed by more careful analysis, and the second can be fixed by more donors.

The difference between the tuned and the frozen pipeline AUC is reported as a column (`frozen_minus_tuned_auc`, Table S4) and is deliberately **not** used as a feasibility rule. A gap between two different algorithms does not establish that a comparison is unanswerable. Every inferential statement in this paper is made about the frozen pipeline, which is the statistic both permutation nulls are built from; the tuned AUC is descriptive.

9.8. Seed stability

All nine comparisons were re-run at five seeds (20260907-20260911), varying the seed for fold construction, permutation draws, and the cross-fitting split used for V_D and its own permutation null. All five per-seed outputs are released rather than summarised (Figure S7, Table S8), and every range quoted in Table 1 and in the text is the minimum and maximum over those five runs. During this work we found a bug: the seed was bound as a default argument at function definition time, so `--seed` never reached the V_D null and the V_D columns came out identical across seeds. The corrected implementation reads the seed at call time. Every number reported here comes from the corrected version.

9.9. Simulation

Cohorts of 200 donors and 100 features were generated. A latent variable z produced a site indicator and a continuous quality proxy; the diagnosis was generated as $\rho \cdot z + \sqrt{(1-\rho^2)} \cdot e$ and then either used continuously or dichotomised. Biological and technical loading vectors overlapped partially (technical = 0.7·random + 0.3·biological, renormalised). Seven regimes were crossed with $\rho \in \{0, 0.25, 0.5, 0.75, 0.9, 0.98, 1\}$ and both label types, at 200 replicates per cell, giving 19,600 simulated cohorts. The seven regimes are:

1. **additive linear**, the base case;
2. **nonlinear design effect**: tanh plus a quadratic term in the quality proxy;
3. **heteroscedastic noise**: the standard deviation depends on the site;
4. **design-by-cell-type interaction**: two feature blocks whose mixing weight depends on the site, with the biological effect confined to one block;
5. **20% case rate, matched thresholds**: the collection boundary is moved to the same quantile as the diagnosis threshold;
6. **20% case rate, offset boundary**: the collection boundary stays at the median, so the threshold and the boundary are offset;
7. **different acquisition in the target**: the external cohort is acquired under a different design mechanism. The two 20% arms are counted separately because they behave differently, and separating them is what

showed the earlier “rare diagnosis blunts the checks” result to be a threshold artefact (Section 3.3).

Critically, the external target reuses the **source’s** biological loading vector. Regenerating it makes every transfer chance-level by construction and tests nothing; only the design mechanism differs between source and target in the shift arm.

A supplementary arm quantifies what admitting a disease-caused covariate does. A severity variable was generated as $1.2\cdot(y - \bar{y})$ + noise and added to the design matrix. Everything else was held fixed.

9.10. Decision-tree walkthrough

Three cohorts were traced through the tree with all evidence recomputed, and every branch value is released in Table S9. Collection strata used exactly the definition in Section 9.6, so that Q2 and Q4 refer to the same partition. Restriction was evaluated in the largest stratum containing at least five donors of each class; where no stratum qualifies, that is reported rather than worked around by coarsening the strata. Matched random subsets of identical size and case/control composition were drawn 20 times from the full cohort for comparison.

9.11. Recovery of GSE135779 design metadata

Batch, collection year, age, sex, race, ethnicity and clinical variables were read from the original Supplementary Table 1b, and per-library sequencing metrics from Supplementary Table 1c [1]. Study names were linked to analysis donor identifiers through the GEO title/accession map; all 44 donors mapped uniquely. The restoration script writes the donor table, batch-by-label, year-by-label and batch-by-year cross-tabulations, and SHA256 hashes of every source file. For GSE174188, per-cell `Processing_Cohort` was read from the distributed CELLxGENE object and reduced to donor fractions over four waves. The dominant wave is the maximum-fraction wave; a pure-wave donor has at least 99% of analysed cells in one wave.

9.12. Representations

Frozen Geneformer. Cell embeddings used Geneformer V2-316M at repository revision `04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5` [4]; checkpoint, configuration, token dictionary and gene-median dictionary hashes are in the methods manifest. Gene identifiers were version-trimmed and mapped to the V2 vocabulary. Within each cell the ranking value was `raw count / total count × 10,000 / Genecorpus-104M gene median`, and genes were sorted descending. Sequences used V2 start/end token IDs 2 and 3, padding ID 0 and maximum length 4096, leaving at most 4094 ranked genes. The model output was the final `last_hidden_state`; token position 0 was kept as the 1,152-dimensional cell vector. The encoder was frozen. Donor representations are coordinate-wise means of sampled cell vectors, sampled without replacement with integer seed 1, capped at 500 cells per donor for GSE135779 and 1,000 for GSE174188 and GSE285773. The embedding environment used Python 3.10, PyTorch 2.5.1 and Transformers 4.46.3.

Pseudobulk. We use pseudobulk rather than a learned latent space as the donor-level representation, so that the comparison between representations is not itself mediated by a model fitted with batch covariates [5]. Stored donor-by-gene values are `log1p(CPM)`. Inside every outer training fold, gene variance was computed on training donors, the 4,000 highest-variance genes were selected, and coordinates were standardised by training-donor mean and standard deviation. HVG pseudobulk uses these values directly. PCA pseudobulk fits up to 30 components on the training values and applies the fitted map to held-out donors. For cross-cohort transfer, trimmed gene identifiers were intersected before source-only HVG selection; feature means, variances, selected genes, scaler, PCA, classifier and regularisation were all fitted on source donors and applied unchanged to the target.

9.13. Fold-contained residualisation and its variants

Unadjusted and residualised models share outer folds, feature construction, scaling, PCA and classifier selection. Inside each outer training fold, design variables were encoded as in Section 9.5 and a multivariate ridge regression (`alpha` 1, intercept fitted and **not** penalised) mapped design to every raw representation coordinate. Training and held-out coordinates were replaced by observed minus predicted values, after which the identical pipeline was fitted to training residuals and applied to held-out residuals. The diagnosis was never given to the residualiser.

Three further arms were run on GSE135779. A multivariate random-forest residualiser (96 trees, depth 4, minimum leaf 3) fitted only on outer-training donors, with pseudobulk genes pre-selected by training-donor variance. A batch-location arm subtracting training-batch mean offsets from training and held-out features; because it applies no empirical-Bayes scale shrinkage we describe it as ComBat-style location adjustment rather than ComBat. And an overlap-weighting arm fitting a training-only logistic diagnosis propensity model from the full design, clipping propensities to 0.025-0.975 and training the representation classifier with overlap weights.

For the central GSE174188 comparison, residualisation loss was paired by split seed. We bootstrapped the 20 paired repeat-level losses 5,000 times and subtracted the locked matched-restriction discrepancy. These intervals quantify split sensitivity conditional on the analysed donors; they are not population confidence intervals.

To probe confound leakage with a nonlinear downstream learner [28], 100 Gaussian simulations used no biological effect, a technical effect of one, and design-diagnosis association 0.75. Random-forest prediction was compared on raw features, globally residualised features and fold-contained residualised features.

9.14. Restriction and matched donor controls

GSE174188 was restricted to dominant wave 4 and separately to pure wave 4. GSE135779 excluded the all-case batch B1 and retained B2 to B6, each containing both labels. The complete internal pipeline was re-run after restriction. For every observed stratum, 20 donor subsets were drawn without replacement from the full cohort with identical donor count and case/control count, each using one prespecified split seed. The observed-minus-matched difference asks whether the stratum is harder than expected from reduced donor number and label composition alone.

9.15. Cell-type composition

The GSE174188 object was restricted to the same CD4-positive alpha-beta T-cell population and 261 donors as the molecular analysis. Per-donor counts were formed for `T4_naive`, `T4_em`, `T4_reg` and all other or unassigned labels. We evaluated raw closed proportions and centred-log-ratio coordinates; for CLR, 0.5 was added to each donor-category count before closure, the donor mean log abundance was subtracted and the final coordinate omitted. Each representation was evaluated with and without log total CD4-cell yield, under the same repeated donor pipeline, with processing wave predicted one against the rest.

9.16. Transfer and learned pooling

Cross-batch analyses trained on waves 2 and 3 and evaluated wave 4, then reversed. Cross-cohort transfer fitted every statistic on the source cohort; target labels were accessed only after prediction. Target AUC intervals used 5,000 case/control-stratified donor bootstrap resamples and the percentile method. Paired AUC comparisons used DeLong tests on identical target donors [23] with Benjamini-Hochberg correction inside declared method families [24].

DeepSets, gated-attention multiple-instance learning and pooling by multi-head attention operated on cap-500 frozen cell embeddings [20-22]. Cell standardisation, a whitened source PCA32 projection, network weights and hyperparameters were all source-fitted. Hidden widths were 16 or 32, weight decay 0.01 or 0.1, and Adam used learning rate 0.02 for 150 epochs; five-fold source-donor cross-validation selected the configuration, three initialisations were fitted on all source donors, and target probabilities were averaged. An ordinary donor mean in the identical PCA32 space isolates the contribution of learned pooling. The PCA32 projection used for hyperparameter selection was fitted once on all source cells before source-donor folds and never used target cells or labels. Source-internal pooling AUC is therefore not an unbiased generalisation estimate, and only independent-target and paired target-donor comparisons support the learned-pooling conclusion.

9.17. Software and reproducibility

The donor-level analyses used Python 3.11-3.13 with NumPy 2.3-2.5, pandas 2.3-3.0, scikit-learn 1.9.0, PyArrow and joblib; exact per-run versions are recorded in each output manifest. Figures were produced in R 4.6.0 with ggplot2 4.0.3 and patchwork 1.3.2. The screen, the extended simulation, the permutation calibration and the decision-tree walkthrough were run on a 16-vCPU ARM container; all other analyses were run locally. Input hashes, seeds, parameter grids, donor-level predictions, complete null distributions and output manifests accompany the analysis.

Random seeds are derived with `hashlib.blake2b` from a recorded master seed and the cell identity. Python's built-in `hash` is not used: it is salted per process for strings, so it would not reproduce across interpreter sessions. The derived seed for every cell is written into the released result tables. No script contains a machine-specific path: the figure scripts locate the project root from their own file position or from `RHEUMLENS_ROOT`, and the analysis scripts take input and output paths as arguments.

Everything downstream of the compute-heavy runs is rebuilt by one command, `bash tools/build_all.sh`: the supplementary tables, Table 1, all fifteen figures, the typeset manuscript, and two checks that fail with a non-zero exit status. The first re-derives numbers quoted in the text from the result tables and requires each to appear verbatim. The second checks the length of the abstract and author summary, that the abstract is unstructured, that figures and tables are cited in the order they are declared, and that each declared item is cited at least once. It also refuses any placeholder, retired term or withdrawn phrase, and it looks inside the rendered figures and the released table headers, where a check on the manuscript text alone cannot see them. The submission package is assembled by a further script from the same outputs, so a figure number cannot differ between the manuscript and the files.

9.18. Use of generative AI

Anthropic Claude (models Claude Opus 4.1 and Claude Opus 5), accessed through the Claude Code command-line interface, was used during this work in the following places and no others.

- **Code review and refactoring** of the analysis scripts in `scripts/cohorts/`, `sim/` and `walkthrough/`. Every statistical definition, gate and threshold in those files was specified by the authors; the model's contribution was implementation, review of implementation, and the identification of two defects that the authors then confirmed by inspection and by rerunning the affected analyses.
- **Figure code.** All figures were drawn by R scripts written with model assistance (`figures/src/*.R`). Every number plotted was read from a released result table, and the correspondence between figure and table was checked by the authors panel by panel.
- **Manuscript language and structure**: rewriting for length and readability, reorganisation of sections, and consistency checking of cross-references and reference numbering.

No text, number, figure or citation was accepted without author verification against the underlying result tables. The model did not select cohorts, choose thresholds, decide which analyses to report, or interpret results. Prompts were not used to generate data of any kind. The authors take full responsibility for the content of this publication.

10. Data and code availability

All cohorts are public. The lupus datasets are at GEO accessions GSE135779, GSE174188 and GSE285773; the COVID-19, influenza and CMV cohorts were obtained through the CELLxGENE Discover curation API and their dataset identifiers are in `cohorts/registry.yaml`. The MIT-licensed repository is at https://github.com/LightChainr/rheumlens with a permanent Zenodo record at https://doi.org/10.5281/zenodo.20813922. The versioned research object accompanying this manuscript contains the cohort registry with API-verified donor counts, the donor-level interface tables, every result table underlying Figures 2 to 8, all five per-seed outputs rather than a summary, the 19,600-cohort simulation output, analysis scripts, locked environments, `REPRODUCE.md` and `SHA256SUMS`. Public raw single-cell matrices remain at their original accessions.

11. Declarations

Author contributions. Conceptualization, H.Y. and D.L.; methodology, H.Y.; software, H.Y.; validation, H.Y. and D.Y.; formal analysis, H.Y.; data curation, H.Y.; writing - original draft preparation, H.Y.; writing - review and editing, D.Y. and D.L.; visualization, H.Y.; supervision, D.L.; project administration, D.L. All authors have read and agreed to the submitted version of the manuscript.

Funding. This research received no external funding.

Competing interests. The authors declare no competing interests.

Ethics approval. Not applicable. This study analysed only publicly available, de-identified single-cell datasets released by their original investigators under the consent and approvals obtained at each contributing site. No new human or animal data were collected.

Informed consent. Not applicable, for the reason given above.

Use of AI tools. See Section 9.18, which states what was used, where, and how the output was checked.

Acknowledgments. The authors thank the investigators of GSE135779, GSE174188 and GSE285773, and the contributors to the COMBAT, Stephenson, Ren and CMV cohorts distributed through CELLxGENE Discover, for releasing donor-level data that made this re-analysis possible.

Figure legends

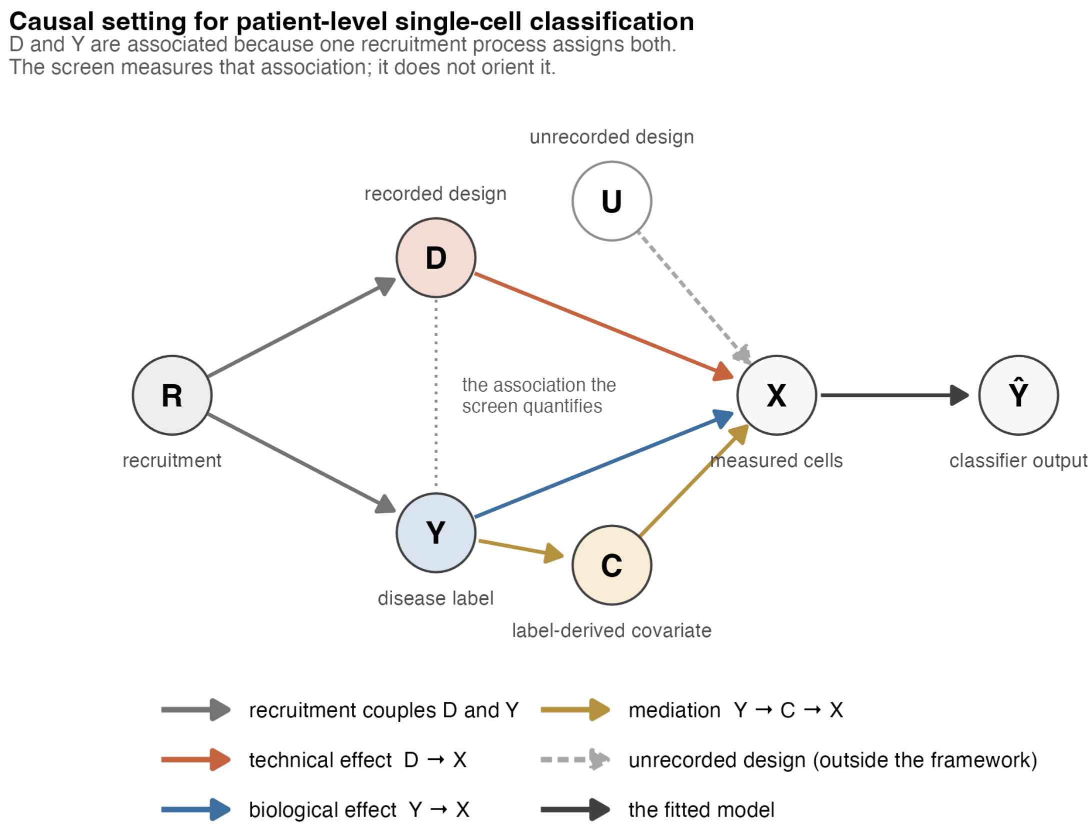

Figure 1. What is being measured. A recruitment process R places each donor in the recorded design D and is also associated with the diagnosis Y; the measured cells X are affected by D (technical) and by Y (biological); a classifier maps X to $\hat{Y}$. C denotes covariates caused by the disease - severity, medication, symptom timing - which lie on $Y \rightarrow C \rightarrow X$ and are excluded from D by blocklist. U denotes collection structure that was never recorded and which nothing in this paper can see. The dotted line marks the association these checks quantify. They measure it; they do not tell you which way it runs.

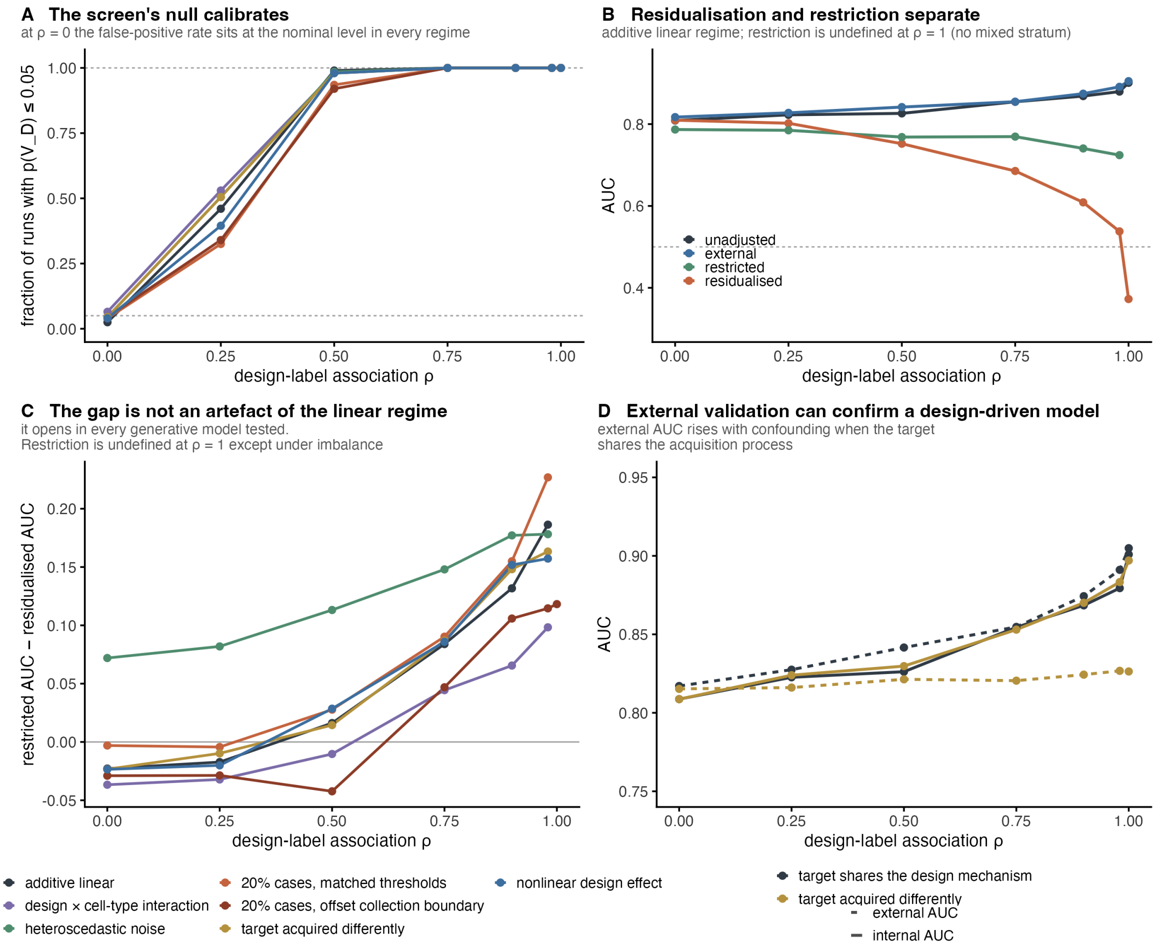

Figure 2. Simulation across seven data-generating regimes. 19,600 simulated cohorts (200 donors, 100 features): seven regimes × seven levels of design-diagnosis association ρ × two label types × 200 replicates. **(A)** Fraction of runs with $p(V_D) \leq 0.05$, binary labels. At $\rho = 0$ this is 0.030 to 0.070 across regimes, bracketing the nominal 0.05. **(B)** Unadjusted, external, restricted and residualised AUC against ρ in the additive linear regime. Restriction is undefined at $\rho = 1$ because no stratum holds both labels. **(C)** Restricted minus residualised AUC for all seven regimes. The gap opens in every one, and a 20% case rate blunts it. **(D)** Internal (solid) and external (dashed) AUC when the target shares the source's collection mechanism versus when it does not. External AUC rises with confounding in the shared case.

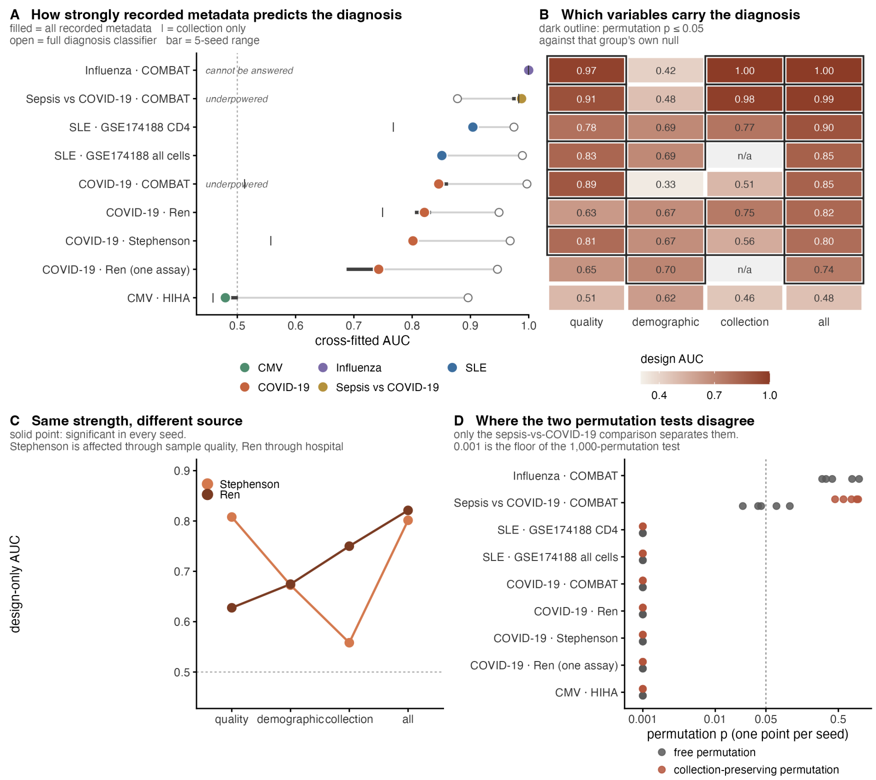

Figure 3. Nine case-control comparisons. (A) Comparisons ordered by cross-validated design-only AUC (filled, coloured by disease) with the corresponding diagnosis-classifier AUC (open) on the same axis. Bars give the range over five random seeds. Dashed line is chance. Italic labels mark the two comparisons that the feasibility check removes from or flags within the ordering. **(B)** Design-only AUC by variable group, rows aligned to (A). A dark outline marks p ≤ 0.05 for the design-only AUC against that group's own permutation null; "n/a" marks a group absent for that comparison. **(C)** The two COVID-19 cohorts with almost equal overall design-only AUC are affected through different variable groups: Stephenson through sample quality, Ren through recruiting hospital. Solid points are significant. **(D)** Free and collection-preserving permutation p-values, one point per seed, for all nine comparisons. Dashed line is 0.05. Only the sepsis-versus-COVID-19 comparison separates the two tests.

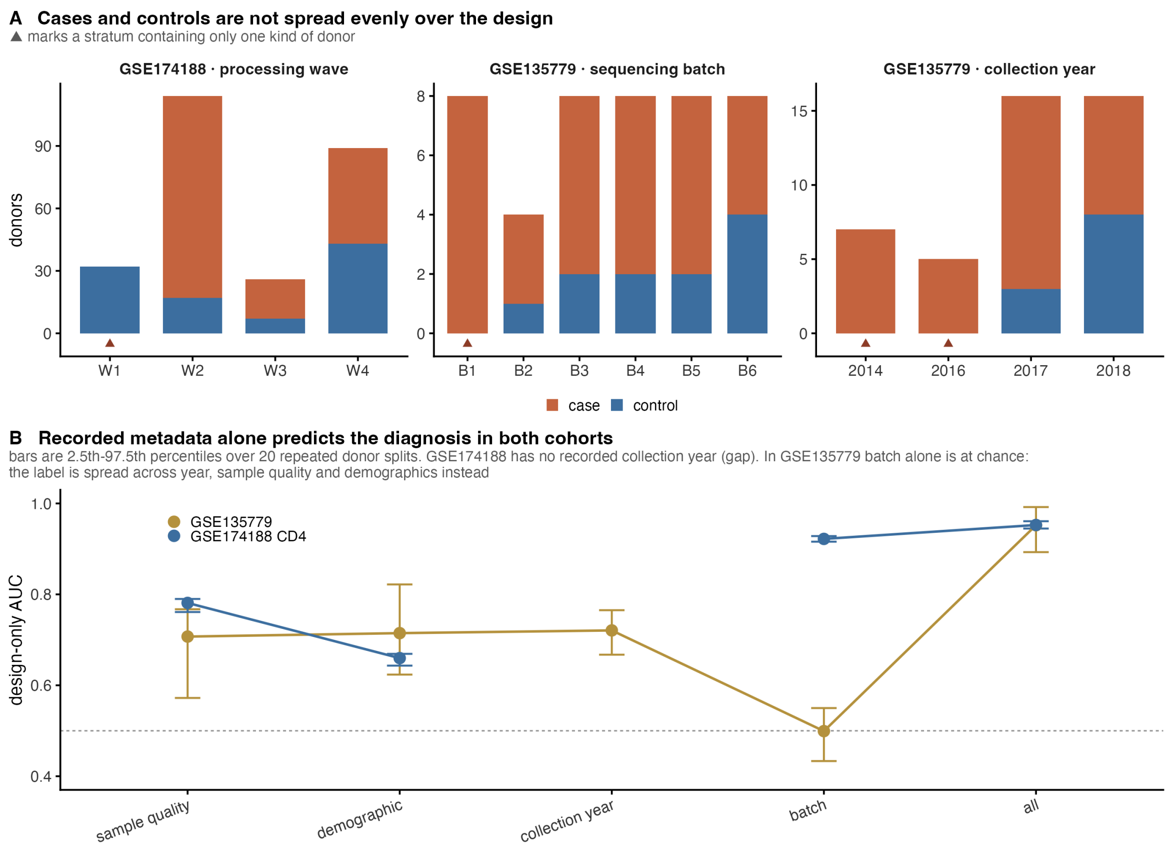

Figure 4. Design and diagnosis in the two lupus cohorts. (A) Donors per collection stratum, split by case and control, for processing wave in GSE174188 and for sequencing batch and collection year in GSE135779. A triangle marks a stratum holding only one kind of donor; such strata cannot be permuted within, and cannot be restricted to. **(B)** Cross-validated AUC for predicting the diagnosis from recorded metadata alone, by variable group, with 2.5th-97.5th percentiles over 20 repeated donor splits. GSE174188 has no recorded collection year, so its line is broken there. In GSE135779 batch alone sits at chance (0.499) while the other groups do not, so the association runs through collection year, sample quality and demographics rather than through batch.

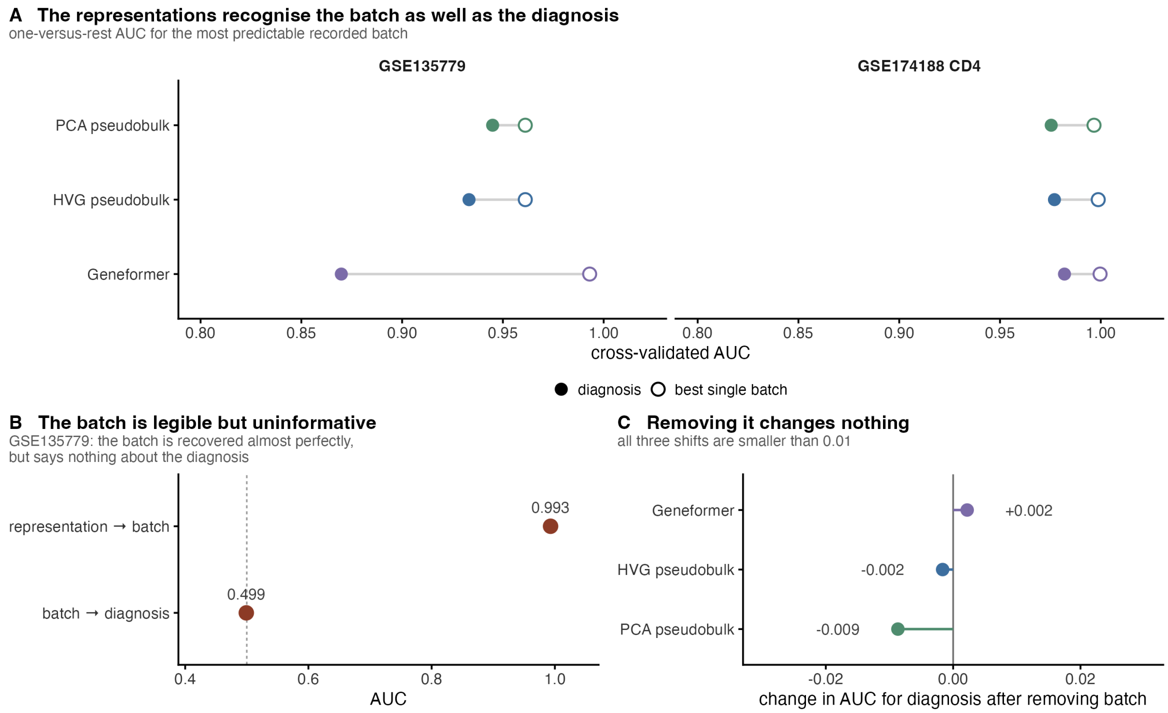

Figure 5. Containing batch information is not the same as using it. (A) For each representation, the cross-validated AUC for the diagnosis (filled) and for the most predictable single recorded batch, one against the rest (open), in both cohorts. Every representation recognises a batch about as well as it recognises the diagnosis. **(B)** GSE135779 negative control. The representation recovers the batch at AUC 0.993, but the batch predicts the diagnosis at 0.499. **(C)** Change in diagnosis AUC after fold-contained batch residualisation in the same cohort: +0.002, −0.002 and −0.009. Batch is legible, uninformative, and its removal changes nothing - so batch predictability on its own supports no conclusion about what the classifier uses.

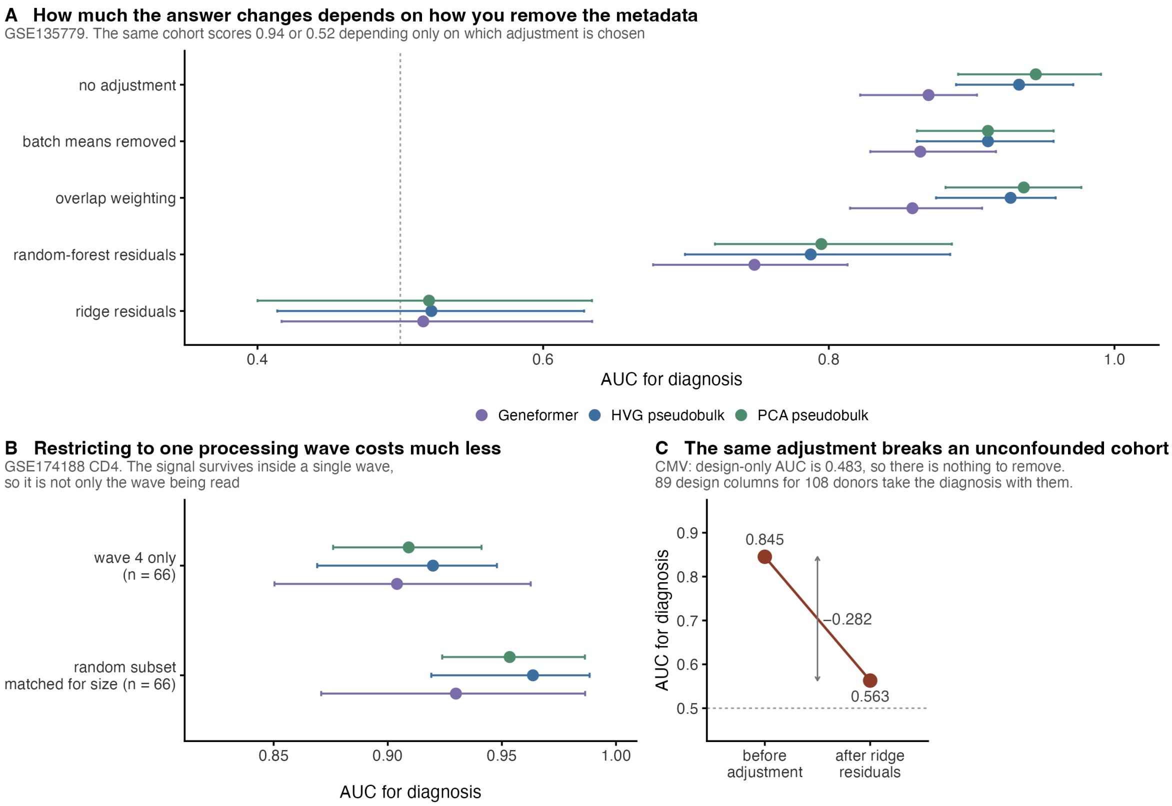

Figure 6. Residualisation and restriction give different answers, and neither is identified. (A) GSE135779 diagnosis AUC under five ways of removing the recorded metadata, for three representations, with 2.5th-97.5th percentiles over 20 repeated splits. The same cohort scores 0.94 or 0.52 depending only on which adjustment is chosen. **(B)** GSE174188 CD4, restricted to the one large near-balanced processing wave (n = 66), against size-matched random donor subsets. The separation largely survives inside a single wave. **(C)** Negative control. In the CMV cohort, where design-only AUC is 0.483 and no design-diagnosis association is detectable, the same ridge residualisation still costs 0.282 AUC. With 89 one-hot design columns for 108 donors, residualisation loss is partly a function of the shape of the design matrix (Figure S2) and cannot be read as a measure of contamination.

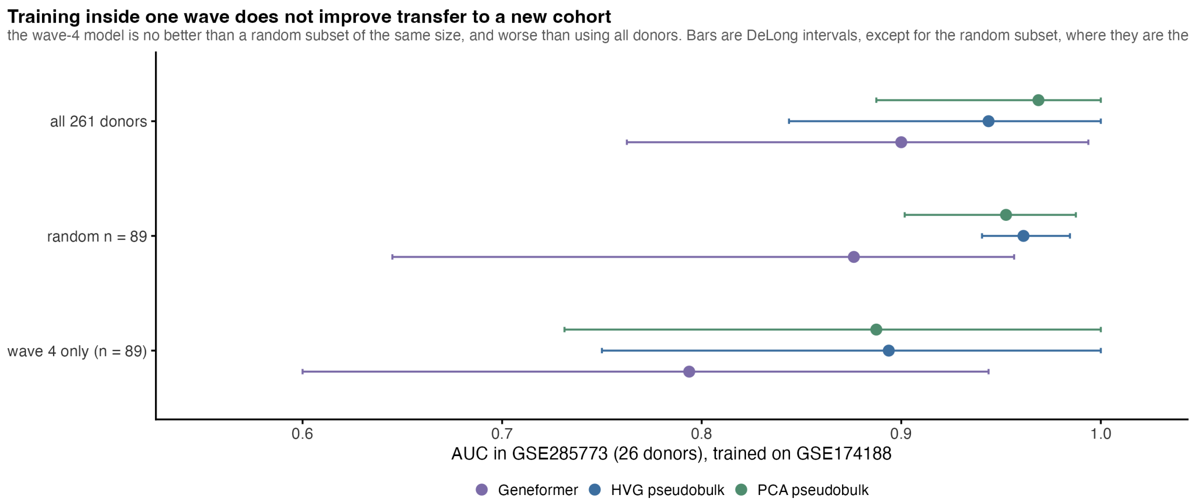

Figure 7. Restricting the training set to one processing wave does not improve transfer. AUC in GSE285773 (26 donors) for classifiers trained on GSE174188, by source set: all 261 donors, a random subset of 89, and the dominant processing wave (n = 89). The wave-restricted source is no better than a random subset of the same size and worse than using every donor. Bars are DeLong intervals, except for the random subset, where they are 2.5th-97.5th percentiles over 20 draws.

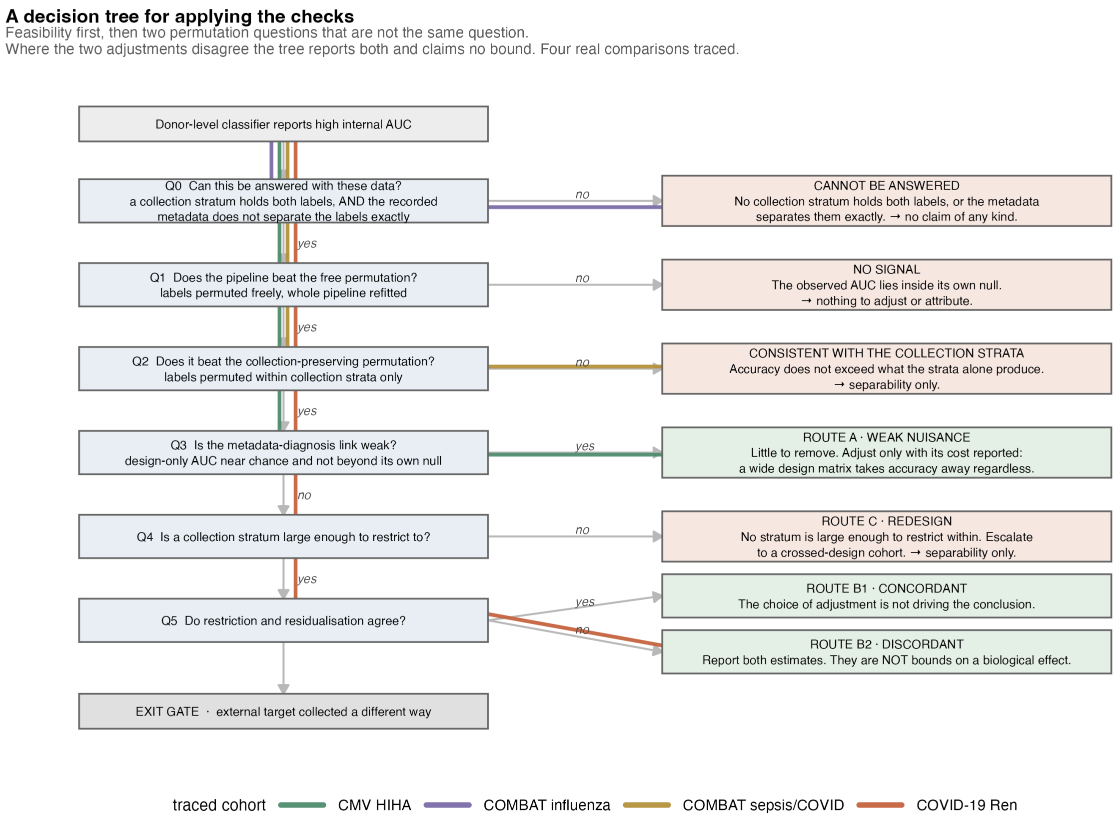

Figure 8. A decision tree for applying the checks. The feasibility question comes first (Q0): a comparison is set aside only when no collection stratum holds both labels, or when the recorded metadata separates the labels exactly. The permutation step is then split into two questions that are not the same one - the free test (Q1) and the collection-preserving test (Q2). Q3 reads the design-only AUC against its own permutation null rather than against a fixed cut-point, and where restriction and residualisation disagree the tree reports both rather than claiming a bound. Coloured paths trace four real comparisons. The strata are the ones the collection-preserving permutation uses, not a separate definition.

Supporting information

Figure S1. Type-I error of the collection-preserving permutation. (A) Fraction of runs rejecting at $p \leq 0.05$ for both permutation tests, as a function of how strongly a sample-quality variable predicts the diagnosis *within* a collection stratum. The generating model contains no disease effect at all, so every rejection is a false positive. **(B)** The mean absolute within-stratum correlation that the restriction leaves in place.

Figure S2. Residualisation loss with no confounding present. AUC lost to fold-contained ridge residualisation in simulated cohorts where the design is independent of the diagnosis by construction, against the width of the design matrix, at 108 donors. The biological effect is calibrated so the unadjusted AUC matches the CMV cohort. The ribbon is the 2.5th-97.5th percentile over 60 draws. The triangle reuses the level sizes of the real CMV batch-by-pool crossing, which is far more ragged than equal blocks. The dashed line marks the loss actually observed in that cohort.

Figure S3. Admitting one disease-caused covariate manufactures the finding. Design-only AUC, V_D, the fraction of runs flagged significant, and residualisation loss, with and without a severity variable added to the design matrix, across five levels of true design-diagnosis association. At $\rho = 0$, where design and diagnosis are unrelated by construction, admitting the covariate raises design-only AUC from 0.488 to 0.793 and flags the cohort in 100% of runs against 5.5% without it.

Figure S4. Design composition of each cohort. Case and control counts by collection stratum for all nine comparisons, showing which strata are label-pure and how many donors sit in mixed strata.

Figure S5. Cell-type composition reproduces the whole pattern. (A) Diagnosis AUC in GSE174188 from four cell-type fractions alone, under four encodings, before and after removing the processing wave. Removing the wave costs about 0.20 AUC in every encoding. **(B)** The same encodings restricted to the dominant wave and to the pure wave, each against a size- and label-matched random donor subset, 20 repeated splits per box. Restriction costs 0.03 to 0.04 AUC on average - far less than removal costs in (A). A four-number biological summary shows the same behaviour as a 1152-dimensional foundation-model embedding, which places the problem in the cohort rather than in the embedding.

Figure S6. Learned pooling never beats averaging the cells. Paired differences in AUC between three learned pooling methods (DeepSets, gated-attention multiple-instance learning, pooling by multi-head attention) and four baselines, in both transfer directions, with paired DeLong intervals. Filled points are significant after Benjamini-Hochberg adjustment; every significant difference is negative.

Figure S7. Seed stability. Per-seed values of design-only AUC and its permutation p, V_D and p(V_D), diagnosis AUC and both permutation p-values, for all nine comparisons at five seeds. The p-value panels use a log axis; 0.005 is the smallest value 200 permutations can return.

Table S1. Calibration study: summary output of both arms, the permutation type-I error of Section 3.4 and the residualisation-width study of Section 3.5.

Table S2. Design manifest, generated directly from the model-matrix construction code: for every comparison and every variable group, the raw columns consumed and the number of expanded columns produced. **Table S2b** lists every column present in a covariate file that enters no block, with the reason. **Table S2c** is a machine check that the manifest reproduces the width of every fitted model matrix (34 of 34 blocks agree). **Table S3.** Simulation summary: all seven regimes × seven ρ × two label types, plus the disease-caused-covariate arm. **Table S4.** Full screen output: every comparison × every variable group, with design-only AUC (linear and random forest), V_D in-sample and cross-fitted, null mean, p(V_D), diagnosis AUC, both permutation p-values and stratum counts. **Table S5.** Learned pooling: per-method target metrics and all paired DeLong tests, both transfer directions. **Table S6.** Cohort registry, generated from the registry file and the donor tables: for all seven datasets, the identifier, citation, the donor count reported by the source, the download size, the case and control labels, the donor, case and control counts actually modelled in each comparison, and how and when each entry was verified. **Table S7.** Hyperparameter grids and fixed settings for every model used. **Table S8.** Seed-stability table, all five seeds, all nine comparisons. **Table S9.** Decision-tree walkthrough: all evidence for every branch, three cohorts.

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